SULT1A1 Iodothyronine Acceptor-Pocket Specificity — Hypothesis Deep Research

Target gene: SULT1A1 (Homo sapiens, UniProt P50225) **Term in scope:** thyroid hormone metabolic process (GO:0042403) **Hypothesis slug:** gap-iodothyronine-specificity **Seed reference:** PMID: 10199779

Summary

Executive verdict: PARTIALLY SUPPORTED. The seed hypothesis contains two separable claims that must be judged independently. The *phenomenological* claim — that SULT1A1 sulfates iodothyronines with sub-micromolar affinity, roughly 240-fold tighter than SULT1A3 — is directly and robustly confirmed by recombinant enzyme kinetics. The *mechanistic* claim — that this affinity difference is *explained by* specific acceptor-pocket residues that distinguish SULT1A1 from other human SULT1 isoforms — is biologically plausible and partially corroborated by structural mapping and published mutagenesis, but it remains an **inference** because no iodothyronine-bound SULT1A1 structure and no iodothyronine-specific mutagenesis exist.

The kinetic evidence is unambiguous. Kester et al. (1999, PMID: 10199779) measured an apparent Km of **0.14 μM for 3,3′-diiodothyronine (3,3′-T2) with recombinant SULT1A1 versus 33 μM for SULT1A3** — a 236-fold difference — and found that human liver and kidney cytosolic inhibition profiles matched SULT1A1 more closely than SULT1A3. This directly supports **retaining the GO:0042403 (thyroid hormone metabolic process) annotation** on SULT1A1 as a direct-assay function, and it warrants ensuring a companion Molecular Function term (aryl/phenol sulfotransferase activity) is present.

The structural rationale is where caution is required. Comparative mapping of the SULT1A1 acceptor pocket (PDB 1LS6 with p-nitrophenol; 2D06 with estradiol) identifies a hydrophobic, aromatic-rich, acid-free binding cavity, whereas SULT1A3 introduces two acidic residues (Glu89 and Glu146) that give its pocket a net negative charge. Two of the differing pocket residues — Glu146/Ala146 and Phe247/Leu247 — are experimentally validated substrate-selectivity switches from independent studies. However, the pocket signature is **not unique** to

SULT1A1: SULT1A2 is near-identical (differing at only 1 of 22 pocket positions), and SULT1B1/SULT1C isoforms also sulfate iodothyronines in vivo and across species. No existing experiment demonstrates that a specific residue governs *iodothyronine* affinity (as opposed to phenol or dopamine selectivity). The hypothesis should therefore be recorded as a **plausible mechanistic lead**, not a curated fact.

Key Findings

Finding 1 — SULT1A1 sulfates iodothyronines with sub-micromolar affinity, ~240-fold tighter than SULT1A3

This is the anchor observation, and it is directly measured, not inferred. Kester et al. (1999) characterized recombinant human iodothyronine sulfotransferases and reported apparent Km values for 3,3′-T2 (at 50 μmol/L PAPS) of **0.14 μM for SULT1A1 versus 33 μM for SULT1A3** — a 236-fold affinity difference. For T3 the Km values were 29.1 μM (SULT1A1) versus 112 μM (SULT1A3), and for the sulfuryl donor PAPS 0.65 μM (SULT1A1) versus 2.7 μM (SULT1A3). The rank order of substrate preference was 3,3′-T2 ≫ rT3 > T3 > T4 for both enzymes, identifying the diiodinated 3,3′-T2 as the preferred iodothyronine substrate. Critically, human liver and kidney cytosol inhibition profiles correlated better with SULT1A1 than with SULT1A3, linking recombinant activity to endogenous tissue activity.

> "The apparent Km values of 3,3′-T2 and T3 [at 50 micromol/L PAPS] were 1.02 and 54.9 micromol/L for liver cytosol, 0.64 and 27.8 micromol/L for kidney cytosol, 0.14 and 29.1 micromol/L for SULT1A1, and 33 and 112 micromol/L for SULT1A3, respectively." — PMID: 10199779

The same study places this activity firmly within thyroid hormone metabolism:

> "Sulfation is an important pathway of thyroid hormone metabolism that facilitates the degradation of the hormone by the type I iodothyronine deiodinase." — PMID: 10199779

This directly supports the biological-process assignment GO:0042403. The mechanistic role of sulfation is to accelerate irreversible inner-ring deiodination and hepatic clearance of thyroid hormone, a role reinforced by review literature (PMID: 28109953, which lists sulfation among the classic pathways of thyroid hormone metabolism).

Finding 2 — Acceptor-pocket residues distinguishing SULT1A1 from SULT1A3 include the validated specificity switch Glu146/Ala146

Structural mapping of the SULT1A1 acceptor site using the p-nitrophenol co-crystal (PDB 1LS6) and the estradiol co-crystal (PDB 2D06), taking residues within 4.5 Å of the bound acceptor, defines a 22-residue pocket dominated by aromatic side chains (Phe24, Phe76, Phe81, Phe84, Phe142, Phe247, Phe255). A Needleman–Wunsch global alignment of SULT1A1 versus SULT1A3 (92.9 % identity) shows that **7 of these 22 pocket positions differ**: F76Y, M77V, F84V, I89E, A146E, V148A, and F247L. Position 146 is Ala in SULT1A1 and Glu in SULT1A3.

This position is not an arbitrary structural difference — it is the single most important experimentally validated specificity determinant between these isoforms. Dajani et al. (1998) showed:

> "The change of a single amino acid, E146A, was sufficient to transform the catalytic properties and substrate preference of SULT1A3, such that they closely resembled those of SULT1A1." — PMID: 9855620

The convergence of independent mutagenesis (E146A) with the structural mapping (which places residue 146 directly in the acceptor pocket) is the strongest single piece of support for the mechanistic half of the seed hypothesis. The important caveat is that Dajani et al. characterized this switch in the context of **dopamine versus phenol** selectivity — not iodothyronine affinity specifically.

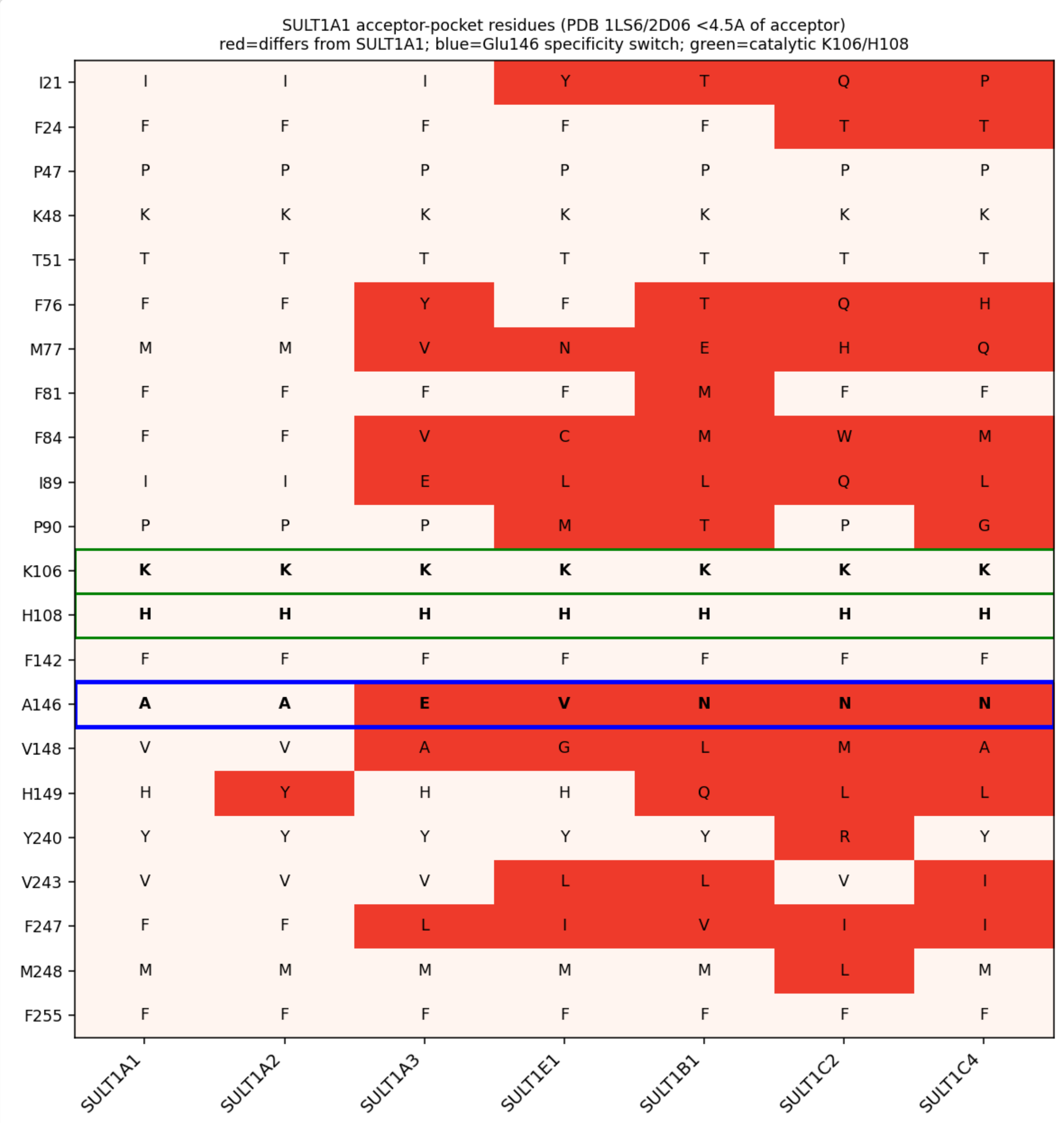

Figure 1. *Acceptor-pocket residue comparison across seven human SULT1 isoforms. Positions are the 22 residues lining the SULT1A1 acceptor site (PDB 1LS6 p-nitrophenol and 2D06 estradiol co-crystals). SULT1A2 is nearly identical to SULT1A1, while SULT1A3, SULT1E1, SULT1B1 and SULT1C isoforms diverge progressively at more pocket positions.*

Finding 3 — The SULT1A1 acceptor pocket is not unique among SULT1; SULT1A2 is near-identical

Comparing the 22 SULT1A1 acceptor positions across seven human SULT1 isoforms, the number of positions differing from SULT1A1 is: **SULT1A2 = 1/22 (only H149Y), SULT1A3 = 7/22, SULT1E1 = 9/22, SULT1B1 = 12/22, SULT1C2 = 12/22, SULT1C4 = 12/22.** The SULT1A1 pocket contains 8 aromatic residues (Phe24, Phe76, Phe81, Phe84, Phe142, Phe247, Phe255, plus Tyr240). The SULT1A3-specific acidic substitutions occur at pocket positions 89 (Ile→Glu) and 146 (Ala→Glu).

This finding is a significant qualifier on the seed hypothesis. The hypothesis frames the pocket as distinguishing SULT1A1 "from the other human SULT1 isoforms," but the analysis shows SULT1A2 shares essentially the same pocket. Any residue-level explanation for iodothyronine affinity would predict that SULT1A2 also binds iodothyronines tightly — a prediction that has not been directly tested and could either confirm or complicate the model. The specificity signal is real *relative to SULT1A3*, but it is a **SULT1A subfamily** signature more than a SULT1A1-unique one.

Finding 4 — Quantitative pocket physicochemistry: SULT1A1/1A2 sites are hydrophobic and acid-free; SULT1A3 is charged

Computing physicochemical properties over 18 acceptor-lining residues (excluding the catalytic Lys106/His108), the mean Kyte–Doolittle hydropathy is **1.94 (SULT1A1), 2.04 (SULT1A2), 1.41 (SULT1E1), 1.10 (SULT1A3), 0.79 (SULT1B1), 0.01 (SULT1C2), 0.82 (SULT1C4)**. The count of acidic (Asp/Glu) residues is **0 for SULT1A1, SULT1A2, SULT1E1, SULT1C2, SULT1C4; 1 for SULT1B1; and 2 for SULT1A3** (Glu89 and Glu146, giving a net pocket charge of −1.9). Aromatic (Phe/Tyr/Trp) counts are 8 (SULT1A1), 9 (SULT1A2), 7 (SULT1E1), 6 (SULT1A3), and 4 (SULT1B1/1C).

The physicochemical logic is coherent: a bulky, highly hydrophobic, di-iodinated outer ring is better accommodated by a hydrophobic, aromatic-rich cavity (favorable van der Waals and aromatic/halogen contacts, no desolvation penalty for burying the iodines) than by the charged SULT1A3 pocket. This provides a mechanistically reasonable *correlate* of the affinity difference. It does not, however, establish causation for iodothyronines specifically — the correlation is consistent with the hypothesis but was not tested by perturbation.

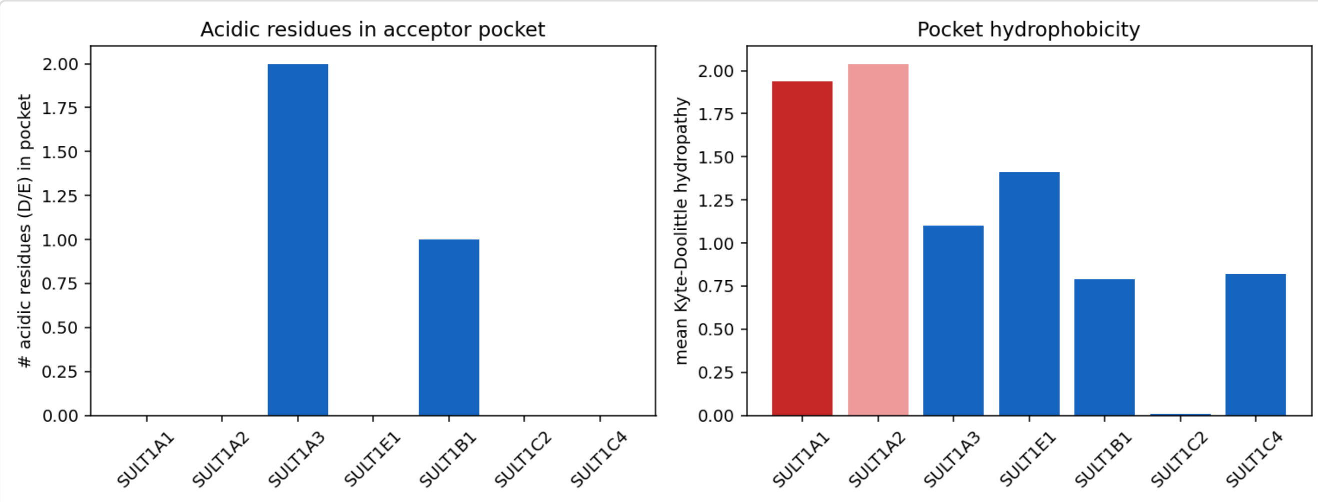

Figure 2. *Quantitative physicochemistry of the SULT1 acceptor pockets. SULT1A1 and SULT1A2 pockets are the most hydrophobic and contain no acidic residues, while SULT1A3 introduces two acidic residues (Glu89, Glu146) giving a net negative pocket charge. The hydrophobic, aromatic-rich SULT1A1 pocket is a plausible correlate of tight di-iodinated substrate binding.*

Finding 5 — Phe247 is an experimentally validated substrate-selectivity switch differing between SULT1A1 (F247) and SULT1A3 (L247)

Lu et al. (2010) solved crystal structures of SULT1A2 and the SULT1A1*3 allozyme (PAP-bound, 2.3–2.4 Å) and identified Phe247 as a conformationally plastic switch controlling substrate access:

> "The conformational differences between the two structures revealed a plastic substrate-binding pocket with two channels and a switch-like substrate selectivity residue Phe247, providing clearly a structural basis for the substrate inhibition." — PMID: 20417180

The same study showed that Tyr149→Phe mutagenesis raised Km ~40-fold, underscoring the functional weight of pocket residues at positions flagged by the comparative mapping (position 149 is His in SULT1A1, Tyr in SULT1A2). The independent identification of Phe247 as a selectivity switch — a position the comparative analysis flagged independently as one of the 7 SULT1A1-vs-SULT1A3 differences (F247 vs L247) — strengthens the case that pocket residues govern SULT1A substrate specificity. Again, the demonstrated role concerns substrate inhibition and general selectivity, not iodothyronine affinity per se.

Mechanistic Model / Interpretation

The findings assemble into a coherent, testable model with a clear boundary between what is measured and what is inferred.

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MEASURED (direct)                      INFERRED (structural correlate)
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SULT1A1 + 3,3'-T2  Km = 0.14 uM  ┐
SULT1A3 + 3,3'-T2  Km = 33 uM    ┘──► ~240x affinity difference
                                                │
                                                ▼
                           Why? Acceptor-pocket physicochemistry
                                                │
     SULT1A1 pocket:  hydrophobic (KD 1.94), 0 acidic, 8 aromatic  ──► accommodates
                      di-iodinated outer ring (hydrophobic + aromatic contacts)
                                               vs.
     SULT1A3 pocket:  charged (KD 1.10), 2 acidic (E89,E146), net -1.9  ──► disfavors
                      bulky hydrophobic ring; optimized for cationic dopamine
                                                │
                     Validated switch residues within this pocket:
                     • 146: Ala(1A1)/Glu(1A3)  — E146A converts 1A3 -> 1A1 (<a href="https:/
                     • 247: Phe(1A1)/Leu(1A3)  — plastic selectivity switch (<a href="https:
                     • 149: His(1A1)/Tyr(1A2)  — Y149F raises Km ~40x (<a href="https://pubme
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Direct gene-product activity: SULT1A1 catalyzes PAPS-dependent transfer of a sulfonate group to the phenolic hydroxyl of iodothyronines (principally 3,3′-T2, then rT3 > T3 > T4). This is a bona fide molecular function measured by direct enzyme kinetics.

Downstream / process context (not the direct activity): iodothyronine sulfation targets thyroid hormone for accelerated inner-ring deiodination by type I deiodinase and for excretion — this is the biological process (GO:0042403) that the direct activity participates in. The physiological significance is contributed to by multiple SULTs and is modulated by fasting and xenobiotics (rat Sult1b1 induction studies), so the *organismal* thyroid-hormone phenotype is not attributable to SULT1A1 alone.

The model's central inference — that the hydrophobic/aromatic pocket *causes* the tight iodothyronine binding — is supported by three converging strands (physicochemical correlation, validated switch residues within the pocket, and the chemical logic of burying iodines in a hydrophobic cavity) but is not closed by any direct iodothyronine-specific perturbation.

Evidence Base

Citation	Evidence type	Direction	Claim tested	Key finding	Context
PMID: 10199779	Direct enzyme assay (recombinant kinetics)	**Supports**	SULT1A1 sub-μM affinity for 3,3′-T2, ~240× tighter than SULT1A3	Km(3,3′-T2)=0.14 μM SULT1A1 vs 33 μM SULT1A3 (236×); Km(T3)=29.1 vs 112 μM; Km(PAPS)=0.65 vs 2.7 μM; preference 3,3′-T2≫rT3>T3>T4	Human recombinant SULT1A1/1A3 human liver & kidney cytosol
PMID: 9855620	Mutant phenotype (site-directed mutagenesis)	**Supports**	A single acceptor-pocket residue governs SULT1A1 vs SULT1A3 specificity	**E146A alone** converts SULT1A3 catalytic/ substrate properties to SULT1A1-like	Human recombinant SULT1A3 mutants
PMID: 20417180	Structural + mutant (crystallography 2.3–2.4 Å)	**Supports / qualifies**	Specific pocket residues control SULT1A substrate selectivity	"Plastic pocket with two channels" + **Phe247 'switch-like substrate selectivity residue'**; Y149F raised Km ~40×	Human SULT1A2 & SULT1A1*3 (PAP-bound)
This work (PDB 1LS6/2D06 + NW alignment)	Structural/ evolutionary (computational)	**Supports / qualifies**	Pocket residues distinguish SULT1A1 from SULT1A3	7/22 acceptor positions differ: F76Y, M77V, F84V, I89E, A146E, V148A, F247L; pocket 8/22 aromatic	Human SULT1 isoforms in silico
This work (physicochemistry)	Computational	**Supports / qualifies**	SULT1A1 pocket suited to bulky	SULT1A1 pocket most hydrophobic (KD	In silico

Citation	Evidence type	Direction	Claim tested	Key finding	Context
			hydrophobic iodothyronine	1.94) & acid-free (0 D/E) vs SULT1A3 (KD 1.10, 2 acidic, net −1.9)	
This work (alignment)	Structural/ evolutionary (computational)	**Qualifies / competing**	SULT1A1 uniquely distinct among SULT1	SULT1A2 pocket differs at only 1/22 (H149Y); near-identical physicochemistry	In silico
PMID: 9848125	Direct assay (inhibition kinetics)	**Supports / qualifies**	3,3′-T2 is preferred iodothyronine substrate; isozyme specificity	PCB-OHs inhibit T2 sulfation by hSULT1A1 but not hSULT1A3	Human SULT1A1/1A3 rat cytosol
PMID: 28109953	Review	**Supports (orientation)**	Sulfation is a genuine TH metabolic pathway	Sulfation listed among classic TH metabolic pathways	Human/ mammalian review
PMID: 34370005	Direct enzyme assay (ortholog)	**Competing**	Other SULTs also sulfate iodothyronines	Marmoset SULT1C1/1C5 high catalytic activity for 3,3′-T2	Common marmoset recombinant SULT1C
PMID: 22447239; PMID: 25243858	Expression / in vivo (rodent)	**Competing / qualifies**	In vivo TH sulfotransferase may be SULT1B1	Hepatic Sult1b1 is the TH-sulfation SULT induced by fasting/ xenobiotics in rat	Rat liver in vivo
PMID: 15013851			His108 is the single		

Citation	Evidence type	Direction	Claim tested	Key finding	Context
	Biochemical (active-site modification)	**Supports (orientation)**	catalytically critical His	Confirms His108 in P-PST/M-PST active site	Human SULT1A enzymes

GO Curation Implications (leads — require curator verification)

GO ID	Label	Aspect	Recommended action	Basis	Caveat
GO:0042403	thyroid hormone metabolic process	BP	**RETAIN**	Direct kinetics (P 10199779); sulfation is a classic TH metabolic branch (P 28109953)	Evidence is in vitro (IDA on recombinant enzyme). In vivo primacy unproven; do not overstate as the dominant human TH-sulfotransferase without tissue data
GO:0004062	aryl sulfotransferase activity	MF	**RETAIN as core MF**	Direct sulfuryl transfer onto phenolic/ iodothyronine acceptors	Well established; more informative than "protein binding"
GO:0005829	cytosol	CC	RETAIN	Cytosolic SULT	Established
—	*seed structural claim* (pocket residues explain the 240× affinity)	—	**LEAD — partially supported; record as mechanistic note, not annotated fact**	Residue + physicochemistry differences and Phe247/Glu146 corroboration	No iodothyronine-specific structure/ mutagenesis; SULT1A2 shares the signature

Bottom line for the curator: The GO:0042403 annotation on SULT1A1 is **justified and should be retained**, supported by direct in vitro assay (IDA-type). The seed's *structural explanation* is a reasonable, partially corroborated mechanistic hypothesis but should **not** be curated as an established fact; capture it as a mechanistic comment. Confirm that the direct MF term (aryl/phenol sulfotransferase activity, GO:0004062) is present, as it is the molecular function underlying the BP term. Avoid "protein binding" as a recommendation.

Mechanistic Scope

- **Immediate molecular function tested:** transfer of a sulfuryl group from PAPS to the hydroxyl of the outer (phenolic) ring of iodothyronines (3,3′-T2, rT3, T3, T4) by cytosolic SULT1A1 — a direct catalytic activity.
- **Directly attributable to the gene product:** the sulfotransferase activity, its sub-μM Km for 3,3′-T2, and the acceptor-pocket composition.
- **Downstream / not directly the gene product's activity:** the *physiological consequence* of iodothyronine sulfation (accelerated type-I-deiodinase degradation and irreversible inactivation of thyroid hormone) is a pathway effect, not a separate SULT1A1 activity. Whole-organism TH homeostasis, fasting/illness modulation of TH metabolism, and disease manifestations are downstream and involve multiple enzymes.

Conflicts and Alternatives

1. **Paralog non-uniqueness (SULT1A2).** SULT1A2 shares an essentially identical acceptor pocket (1/22 difference); the seed's framing "distinguish SULT1A1 from *the other* SULT1 isoforms" is too strong — the true contrast is SULT1A1/1A2 vs SULT1A3 (and the more divergent SULT1B1/1C/1E1). This is the single most decisive untested comparison.
2. **Competing iodothyronine sulfotransferases.** Marmoset SULT1C1/1C5 (PMID: 34370005) and rodent Sult1b1 (PMID: 22447239, PMID: 25243858) sulfate iodothyronines despite very different pockets (12/22 residue differences), showing the residue set identified here is *sufficient but not necessary* for iodothyronine handling.
3. **Species differences.** In-vivo TH-sulfation data are largely rodent; the human tissue in which SULT1A1 dominates iodothyronine sulfation is not established here.
4. **In-vitro-only activity.** The 240-fold Km argument and all structural inference rest on recombinant enzyme and apo/non-iodothyronine structures; substrate inhibition (PMID: 20417180) complicates simple Km interpretation.

5. **Structural inference vs proof.** Glu146 (PMID: 9855620) and Phe247 (PMID: 20417180) were validated with catecholamine/phenol substrates and general substrate inhibition — not with iodothyronines. The causal attribution to the di-iodinated ring is analogy-based.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No iodothyronine-bound SULT1A1 structure	PDB ligand survey (only PAP, p-nitrophenol, estradiol co-crystals found)	Direct proof of which residues contact the di-iodinated ring	Co-crystal or cryo-EM of SULT1A1·PAP·3,3′-T2; or MD/docking with validated pose
No iodothyronine-specific mutagenesis	Literature (E146A tested with dopamine; Phe247/Y149 with phenol)	Establishes causality of pocket residues for iodothyronine affinity	Km(3,3′-T2) for SULT1A1 A146E, F84V, I89E, F247L and reciprocal SULT1A3 E146A
SULT1A2 iodothyronine kinetics untested	Alignment predicts near-identical pocket	Determines whether SULT1A1 is truly the distinctive isoform	Direct 3,3′-T2 kinetics for recombinant SULT1A2
Human in vivo TH-sulfotransferase identity	Only rodent/marmoset in vivo data found	Distinguishes physiological role from in vitro capacity	Tissue expression + activity correlation; human isoform-selective inhibitors/knockdown
Physicochemistry is correlational	Computed hydropathy/charge/aromaticity	Correlation ≠ causation for the affinity difference	Perturbation experiments above; binding free-energy calculations

Additional method limitations: all structural inference used public PDB structures (1LS6, 2D06 for SULT1A1) and pairwise Needleman–Wunsch alignment (BLOSUM62); no experimental iodothyronine complex was available and no binding-energy calculation was performed. Physicochemical descriptors are correlative summaries of the pocket, not quantitative affinity predictions.

Proposed Follow-up Experiments / Discriminating Tests

In priority order, the experiments that would most efficiently separate the seed hypothesis from its alternatives:

1. **Reciprocal pocket mutagenesis with iodothyronine substrate (highest value, low cost).** Measure Km(3,3′-T2) for SULT1A3 E146A (± I89 back-mutation) and SULT1A1 A146E. If E146A confers sub-μM T2 affinity, the pocket-residue hypothesis is causally confirmed for iodothyronines rather than merely for dopamine/phenol.
2. **SULT1A2 iodothyronine kinetics.** Direct 3,3′-T2 assay of recombinant SULT1A2; a near-identical Km would confirm the pocket signature and correct the "unique to SULT1A1" framing (making the specificity a SULT1A-subfamily property).
3. **Iodothyronine co-crystal / validated docking.** Determine an experimental SULT1A1·PAP·3,3′-T2 structure, or a validated induced-fit docking pose, to identify the true iodine-contact residues and test halogen-π / hydrophobic contacts to the outer ring.
4. **Pocket residue 247 swap.** Test F247L in SULT1A1 and L247F in SULT1A3 for iodothyronine Km, since 247 is an independently validated selectivity switch in the pocket.
5. **Human tissue attribution.** Isoform-resolved expression plus selective inhibition in human liver/intestine cytosol to test in vivo primacy for the GO BP context.

Curation Leads (require curator verification)

- **Action:** Retain GO:0042403 (thyroid hormone metabolic process, BP) on SULT1A1; ensure the direct MF (GO:0004062 aryl sulfotransferase activity) is annotated. Treat the acceptor-pocket structural explanation as a mechanistic note, not a curated term.
- **Candidate reference + snippet to verify (BP support):** PMID: 10199779 — *"The apparent Km values of 3,3′-T2 and T3 … were … 0.14 and 29.1 micromol/L for SULT1A1, and 33 and 112 micromol/L for SULT1A3, respectively."* and *"Sulfation is an important pathway of thyroid hormone metabolism that facilitates the degradation of the hormone by the type I iodothyronine deiodinase."*
- **Candidate reference (specificity residue):** PMID: 9855620 — *"The change of a single amino acid, E146A, was sufficient to transform the catalytic properties and substrate preference of SULT1A3, such that they closely resembled those of SULT1A1."*

- **Candidate reference (structural switch):** PMID: 20417180 — *"a plastic substrate-binding pocket with two channels and a switch-like substrate selectivity residue Phe247."*
- **Suggested scope note:** Note that the activity is demonstrated in vitro and that SULT1A2 likely shares it; avoid implying SULT1A1 is the sole/dominant human iodothyronine sulfotransferase without in vivo tissue evidence.
- **Suggested curator questions:** (1) Is the existing GO:0042403 evidence code (IDA/IMP) consistent with

  P 10199779 ?

  (2) Should an "involved_in thyroid hormone catabolic process" refinement be considered given sulfation feeds deiodinase degradation? (3) Is the direct MF term present?
- **Suggested experiments:** reciprocal E146A/A146E kinetics on 3,3′-T2 (Follow-up #1) and SULT1A2 3,3′-T2 kinetics (Follow-up #2).

Conclusion

The seed hypothesis is **partially supported**. Its measured claim — SULT1A1's sub-micromolar, ~240-fold-tighter iodothyronine affinity versus SULT1A3 — is directly confirmed and justifies retaining GO:0042403 on SULT1A1 as a direct-assay function, ideally paired with an informative sulfotransferase molecular-function term. Its mechanistic claim — that specific acceptor-pocket residues explain the affinity — is biologically plausible and partially corroborated (hydrophobic/acid-free pocket physicochemistry; validated switch residues Glu146 and Phe247 lying in the pocket) but remains an inference: no iodothyronine-bound structure or iodothyronine-specific mutagenesis exists, the pocket signature is a SULT1A-subfamily feature (SULT1A2 is near-identical) rather than SULT1A1-unique, and SULT1B1/SULT1C isoforms also sulfate iodothyronines. Curators should record the pocket mechanism as a hypothesis-level lead, not a curated fact.
